



Ensemble Models & Random Forest Regression



Data Analytics Lab

dal.global@grupo.cefet-rj.br | <https://eic.cefet-rj.br/dal>

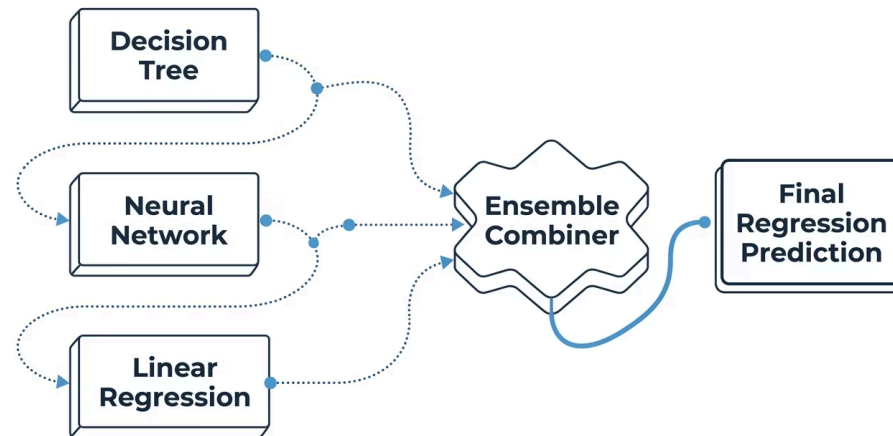


Introduction to Ensemble Models for Regression

Ensemble methods in regression combine predictions from multiple individual models to achieve superior predictive performance compared to any single model.

This approach effectively reduces variance, improves accuracy, and captures complex non-linear relationships in data more robustly.

We will explore key techniques such as Random Forest, Gradient Boosting, and various model averaging strategies.



Ensemble models leverage the strengths of diverse predictors to output a more reliable and accurate forecast.

Random Forest: Theoretical Foundations

Random Forest is an ensemble method that combines multiple decision trees trained independently. Each tree is built on a bootstrap sample of the data and uses a random subset of variables at each split, making the model robust and low-variance.

Bagging (Bootstrap Aggregating)

Each tree is trained on a random sample *with replacement* from the training set. This ensures diversity among trees and reduces overfitting.

Random Variable Selection

At each split node, only a random subset of `mtry` variables is considered. This decorrelates the trees and further increases the diversity of the ensemble.

Variance Reduction

Averaging the predictions of many decorrelated trees reduces the variance of the final model without increasing bias, improving generalization.

Final Prediction

For regression, the final prediction is the **arithmetic mean** of the predictions from all `ntree` trees in the forest.

Main Hyperparameters

`mtry`

Number of variables randomly selected to evaluate at each node split. Smaller values increase diversity among trees; larger values bring the model closer to a single tree. Default recommendation: $p/3$ for regression.

`ntree`

Total number of trees in the forest. More trees increase prediction stability, but with growing computational cost. Typical values: 100 to 500 trees.

References: Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. | Liaw, A. & Wiener, M. (2002). Classification and Regression by randomForest. *R News*, 2(3), 18–22.

reg_rf: Random Forest for Regression

Setup & Data

```
# Regression Random Forest

library(daltoolbox)
library(MASS)

data(Boston)
Boston <- as.matrix(Boston)

set.seed(1)
sr <- sample_random()
sr <- train_test(sr, Boston)
boston_train <- sr$train
boston_test <- sr$test

# mtry: variables per split
# ntree: number of trees
model <- reg_rf("medv", mtry=7, ntree=30)
model <- fit(model, boston_train)
```

Training & Evaluation

```
# Training evaluation
train_pred <- predict(model, boston_train)
train_eval <- evaluate(model,
  boston_train[, "medv"], train_pred)
print(train_eval$metrics)
# mse=1.36, smape=0.04, R2=0.98

# Test evaluation
test_pred <- predict(model, boston_test)
test_eval <- evaluate(model,
  boston_test[, "medv"], test_pred)
print(test_eval$metrics)
# mse=14.60, smape=0.12, R2=0.76
```

Step-by-Step

- Define the model with `reg_rf("medv", mtry=7, ntree=30)`, where `medv` is the target variable
- `mtry`: number of variables considered at each split
- `ntree`: number of trees in the ensemble
- Training $R^2=0.98$ indicates a near-perfect fit — watch for overfitting
- Test $R^2=0.76$ is the best generalization result achieved so far

📌 **Key Concept:** Random Forest averages many trees trained with randomness, effectively **reducing variance**. The gap between training and test performance (0.98 vs. 0.76) is expected and typical of ensemble methods. Ref: Breiman (2001).

Acknowledgements



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